MILITARY SPECIFICATION

WRENCH, COMBINATION BOX AND OPEN END, HIGH STRENGTH, THIN WALL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the combination wrenches described herein shall consist of this document and the latest issue of MIL-W-8982.

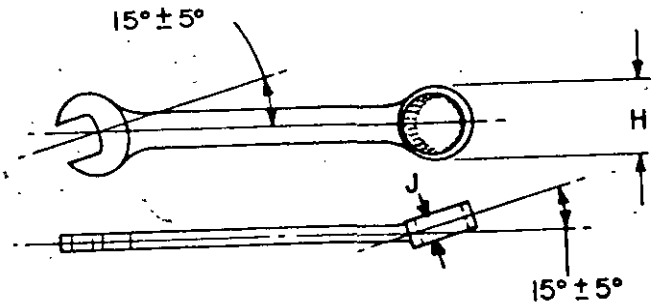


FIGURE 1. Combination wrench.

REQUIREMENTS:

1. The combination wrench shall conform to figure 1, table I, and the applicable portions of ANSI B107.6. Spline and chamfer dimensions shall conform to MS33787. This wrenching design fits fasteners with the corresponding MS33787 spline as well as bihexagonal and hexagonal wrenching configurations.

2. The combination wrench shall be hardened to 45-55 HRC.

3. The combination wrench shall be chromium plated in accordance with QQ-C-320, Class 1, Type I. Cadmium plating is not permitted.

4. The combination wrench shall withstand the endurance torque and ultimate torque loads specified in table I. There shall be no damage or permanent deformation (set) to the wrench.

5. The wrench shall be engaged with hexagon and spline mandrels in accordance with table I. It shall be subjected to the endurance torque load and ultimate torque load values specified in table I for 10 seconds. The size of the hexagon mandrels in the engaged portion shall be equal to (+ 0.000, - 0.002) the basic wrench size. The spline mandrel dimensions in the engaged portion shall be in accordance with MS33787, table I, except that C and W diameters held to a minimum dimension (+ 0.002, - 0.000). Mandrels shall be hardened to not less than 55 HRC and smoothly finished in engagement area of wrench openings.

AMSC N/A FSC 5120

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TABLE I. Combination wrench dimensions and box end test loads.

Dash No.	Basic Wrench Size	Outside Diameter H Max. (in.)	Thickness J Max. (in.)	Spline Test			Single Hex Tests		
				Ultimate Torque (in. -lbs)	Endurance Torque (in. -lbs)	Engage Length (in.)	Ultimate Torque (in. -lbs)	Endurance Torque (in. - lbs)	Engage Length (in.)
06	0.188	0.320	0.187	160	95	0.091	95	60	0.125
07	0.218	0.359	0.203	235	140	0.110	135	85	0.125
08	0.250	0.400	0.219	300	195	0.125	175	115	0.125
09	0.281	0.447	0.235	365	220	0.140	210	130	0.141
10	0.312	0.490	0.250	450	250	0.156	300	175	0.156
12	0.375	0.590	0.343	735	515	0.188	400	280	0.188
14	0.438	0.680	0.375	1,250	875	0.219	750	525	0.250
16	0.500	0.745	0.375	2,250	1,575	0.250	1,200	840	0.313
18	0.562	0.825	0.406	3,000	2,275	0.281	1,650	1,150	0.375
20	0.625	0.907	0.468	3,250	2,450	0.312	2,200	1,450	0.438
22	0.688	0.996	0.516	4,250	2,975	0.344	2,600	1,820	0.500
24	0.750	1.090	0.531	4,500	3,500	0.375	3,000	2,100	0.500
26	0.812	1.170	0.547	6,300	4,410	0.406	3,300	2,300	0.562
28	0.875	1.265	0.563	7,150	5,000	0.438	3,600	2,520	0.562
30	0.938	1.375	0.581	8,800	6,160	0.469	4,500	3,150	0.625
32	1.000	1.460	0.594	11,000	7,700	0.500	5,400	3,780	0.750
34	1.062	1.600	0.670			0.531	6,000	4,200	0.750
36	1.125	1.680	0.670			0.563	6,500	4,550	0.875
38	1.188	1.770	0.800			0.594	6,500	4,550	0.875
40	1.250	1.930	0.860			0.625	6,500	4,550	0.875
42	1.312	2.000	0.920			0.656	10,000	7,000	1.000
44	1.375	2.050	0.920			0.688	12,000	8,400	1.000
46	1.438	2.100	0.920			0.719	12,000	8,400	1.000
48	1.500	2.240	1.020			0.750	13,000	9,100	1.125
50	1.562	2.300	1.020			0.781	13,000	9,100	1.125
52	1.625	2.440	1.020			0.813	13,000	11,900	1.125
54	1.688	2.530	1.110			0.844	17,000	11,900	1.250
56	1.750	2.580	1.110			0.875	18,000	12,600	1.250
58	1.812	2.700	1.140			0.906	19,000	13,300	1.250
60	1.875	2.740	1.140			0.938	19,000	13,300	1.250
62	1.938	2.780	1.140			0.969	19,000	13,300	1.250
64	2.000	2.900	1.300			1.000	20,000	14,000	1.375
66	2.062	2.940	1.300			1.031	20,000	14,000	1.375
68	2.125	3.180	1.330			1.062	20,000	14,000	1.375
70	2.188	3.270	1.330			1.094	20,000	14,000	1.500

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6. The wrench shall be subjected to not less than 500 cycles on both the hexagon and spline mandrels in accordance with the endurance torque load and ultimate torque load values specified in table I. Engagement lengths shall be the same as the ultimate tests.

NOTES:

1. Dimensions are in inches.

2. The wrenches shall be of uniform quality. They shall be free from nicks, splits, cracks, laps, seams and scale. They shall have no imperfections that will adversely affect serviceability, reliability or safety.

3. Example of a military part number:

M8982/6 - 06

- 06 — Dash number in thirty seconds of an inch
- M8982/6 — Basic specification sheet number

Custodians:
Army - AV
Air Force - 11
Navy - AS

Review:
Army - GL, AR
Air Force - 99

Preparing Activity:
Navy - AS

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